

BIRENDRA SINGH

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2497/3 West Rathiya, Achhnera, Agra, Uttar Pradesh, India

PROFESSIONAL SUMMARY

Results-driven Computer Science Engineer with a robust foundation in Artificial Intelligence, Machine Learning, Web Development, and Automation. Passionate about architecting data-driven solutions and deploying scalable web and mobile applications. Proven track record of leveraging AI-augmented methodologies (vibe coding) to accelerate development, with deep expertise in predictive modeling, prompt engineering, and cloud infrastructure integration.

KEY COMPETENCIES & TECHNICAL SKILLS

- Programming Languages:** Python, C, C++, SQL, JavaScript
- App & Web Development:** Android SDK, Jetpack Compose, HTML5, CSS3, Django, React
- AI, ML & Automation:** Deep Learning, Computer Vision, Predictive Modeling, Prompt Engineering, n8n Automation
- Tools & Cloud:** Git, VS Code, PyCharm, Linux, Streamlit, Vercel, Render, Supabase, Oracle Cloud (OCI)

RESEARCH & PROJECTS

Real Time Sign Language Recognition System | *Research Paper* | GL BAJAJ University 2026

- Designed an intelligent computer vision system leveraging Deep Learning algorithms to translate sign language gestures into text in real-time.
- Authored and presented the technical research paper, proposing solutions to bridge communication accessibility barriers.

Swara Music | *Native Android Music Platform* | [Live Link](#) 2026

- Pioneering a native Android music streaming platform utilizing Kotlin, Jetpack Compose, and AI-augmented workflows.
- Integrated Android Media3 ExoPlayer for robust background audio streaming and seamless state management.

YouTube Automation Engine | *n8n & Python Integration* | [Live Link](#) 2025

- Engineered an automated, low-code workflow utilizing n8n to heavily streamline the YouTube content pipeline.
- Automated video processing, synthesized AI-generated metadata (titles/tags), and scheduled API-based uploads.

Universal Analyst Model (UAM) v2 | *Full Stack AI Platform* | [Live Link](#) 2025

- Architected a comprehensive data analytics platform providing automated Exploratory Data Analysis (EDA), interactive charts, and dynamic predictive model generation.

WORK EXPERIENCE

NIELIT Jun 2025 – Jul 2025 *AI & ML Intern* Online

- Executed intensive AI/ML pipelines using Python, focusing on optimizing data preprocessing structures.
- Gained hands-on experience in feature engineering, model training, and algorithmic problem-solving techniques.

AI Academia May 2025 *Intern* Hybrid

- Collaborated within an Artificial Intelligence Workshop on developing basic neural network concepts.
- Evaluated and integrated modern ethical AI deployment guidelines into practical minor projects.

EDUCATION & CERTIFICATIONS

EDUCATION

Bachelor of Technology (B.Tech) - CS
Raja Balwant Singh Eng. Technical Campus, Agra
2023 – 2026 | CGPA: 7.0 (Till 7th Semester)

Bachelor of Science (B.Sc)
2021 – 2023 | 53%

Intermediate Education (12th)
NC Educational Academy
2016 – 2018 | 54%

High School (10th)
NC Educational Academy
2014 – 2016 | CGPA: 8.6

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025**
AI Foundations Associate
- **ChatGPT Prompt Engineering**
DeepLearning.AI
- **Python Zero to Hero**
Guvi
- **Artificial Intelligence Workshop**
AI Academia